

1. 5 Concepts I found interesting

I. Dynamic Nature of Visual Receptors

Whereas digital sensors record a steady input stream, rods and cones become adapted very rapidly to the constant stimulus, leading to a loss of recognition of the fixed visual stimulus. In order to prevent sensory adaptation and renew the visual stimulus, there is need for continuous and involuntary microscopic eye movements including microsaccades, drift, and tremor.

II. Image Data Format

Digital images have a mathematical structure of a numeric matrix. For efficiency reasons, multidimensional image data (tensors) e.g., H, W, C, needs to be converted to a 1D array within a computer's memory. This is normally achieved through special formats such as HWC or CHW.

III. Image Resizing

During manipulation with the grid of an image, fractional values are computed either through replicating the nearest-pixel value (nearest neighbor) or averaging pixel values of the surrounding four pixels (bilinear interpolation).

IV. Visual Depth

Although the human eye is capturing 2D representations of the external world, the brain perceives a 3D view through its remarkable ability to compute environment-specific variables such as shadows, color gradients, focus blur, disparity from two eyes, and perspective.

V. RGB HSV Color Space

Computer vision systems analyze raw pixel RGB values directly. But RGB only consists of a small triangular space of all colors Humans can see, for that several other frameworks exist like CMYK. Alternative frameworks like the HSV (Hue, Saturation, Value) color space are often utilized to isolate color properties more effectively than standard RGB but still offer the same amount of colors as RGB.

2. Medical Imaging Reports

Medical imaging is the use of technology to be able to see and understand the inside of the human body for diagnosis, monitoring, and treatment of diseases or injuries.

Types of medical imaging:

1. X-Ray
2. Computed Tomography (CT) Scan
3. Magnetic Resonance Imaging (MRI)
4. Ultrasound
5. Blood Tests
6. HistoPathology

